

Pravallika Modugu

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PROFESSIONAL SUMMARY

Results-driven Data Engineer and Analyst with 4+ years of experience leveraging SQL, Python, Power BI, Tableau, and cloud technologies (AWS, Azure, Snowflake) to transform complex data into actionable business insights. Skilled in data modeling, ETL automation, statistical analysis, and dashboard development, delivering scalable analytical solutions that enhance performance, forecasting, and decision-making. Adept at ensuring data quality, governance, and compliance while collaborating across teams to drive data-informed strategies that improve reporting accuracy, operational efficiency, and overall business growth.

WORK EXPERIENCE

Capital One, USA | Data Engineer

Sep 2024 – Present | Remote

- Engineered automated pipelines using Kafka, Spark, and Airflow on Azure Synapse and Snowflake to process 25M+ daily records, enabling analytical dashboards that optimized fraud detection accuracy and operational reporting efficiency across multiple business units.
- Designed and maintained Power BI and Tableau dashboards displaying financial KPIs, customer metrics, compliance insights, empowering leadership to identify trends, reduce latency in decision-making, and streamline resource utilization through precise, data-driven analysis.
- Developed advanced SQL scripts and Python models for anomaly detection, data transformation, and ad-hoc analysis, reducing manual reporting time by 40% while ensuring data consistency, validation accuracy, and stakeholder confidence in enterprise reporting systems.
- Collaborated cross-functionally with compliance, finance, and product teams to standardize reporting frameworks, improve data lineage and quality assurance processes, align analytics delivery with regulatory standards including SOX, GDPR, internal governance policies.

University at Buffalo, USA | Research Associate – Data Science

Feb 2024 – Aug 2024 | Buffalo, USA

- Performed statistical analysis and predictive modeling using Python (Pandas, NumPy, SciPy) and SQL, improving model accuracy by 15% and generating evidence-based insights to support institutional decision-making and long-term strategic planning.
- Built automated Power BI dashboards to visualize academic performance trends and research outcomes, reducing manual reporting time by 35% while increasing visibility and accessibility of analytical results for faculty and leadership teams.
- Applied regression, hypothesis testing, and causal inference frameworks using DoWhy and EconML to evaluate resource effectiveness and optimize program funding, improving operational efficiency and transparency by 20% across university departments.
- Presented technical findings and visual analytics to non-technical stakeholders, ensuring actionable understanding of data insights while strengthening institutional governance, reproducibility, and policy compliance standards across multiple research initiatives.

Accenture, India | Data Engineer

Jan 2020 – Jun 2023 | Hyderabad

- Designed and deployed real-time data pipelines using Kafka, Spark, and Airflow to process millions of financial records, enabling dynamic dashboards and reports that improved accuracy, reliability, and reduced manual report generation time by 30%.
- Developed Power BI and Tableau dashboards connected to Snowflake and Redshift datasets, transforming raw operational data into visual stories that improved forecasting precision, compliance tracking, and performance monitoring for high-impact business stakeholders.
- Optimized SQL-based ETL scripts, database schemas, and materialized views to accelerate query execution by 25%, enhancing analytical performance, reducing compute costs, and enabling faster, scalable reporting delivery across enterprise data ecosystems.
- Automated multi-source data ingestion using AWS Glue and Lambda to integrate structured and unstructured data from APIs and cloud services, ensuring validated, analytics-ready datasets for accurate visualization, KPI reporting, executive performance review dashboards.

Informative Web Solutions, India | Data Engineer Intern

Jul 2019 – Dec 2019 | Remote

- Assisted in building automated ETL pipelines using Python, SQL, and Airflow to extract, transform, and load diverse datasets, supporting analytics teams with timely, validated, and high-quality data for reporting and decision-making accuracy.
- Collaborated with senior engineers to design Power BI and Tableau dashboards that delivered real-time insights into customer engagement trends, enabling marketing teams to track KPIs, optimize campaign performance, and drive evidence-based business strategies.
- Validated and tested data models within PostgreSQL and Snowflake environments to ensure compliance with enterprise reporting standards, improving reproducibility, data consistency, long-term reliability across analytics pipelines, dashboard development workflows.

EDUCATION

M.S. in Data Science, University at Buffalo, NY

GPA: 3.4 / 4.0

Coursework: Data Analytics, Distributed Systems, AI/ML, Cloud Architecture

SKILLS

- Programming & Analytics:** Python (Pandas, NumPy, Matplotlib), SQL (Advanced Queries, Joins, Window Functions), R, and Excel (Power Query, Pivot Tables, Advanced Formulas) for data analysis, automation, and reporting.
- Visualization & Business Intelligence:** Power BI, Tableau, and Excel Dashboards for KPI reporting, data storytelling, and trend visualization, delivering actionable insights and executive-ready analytical dashboards.
- Data Engineering & Cloud:** ETL and pipeline automation using Airflow, Kafka, Spark, AWS Glue, Databricks, Snowflake, Redshift, and Azure Synapse to integrate, model, and optimize analytics-ready datasets across hybrid cloud environments.
- Governance & Collaboration:** Data quality, lineage, and compliance (SOX, GDPR) using Great Expectations and Apache Atlas, with strong communication, stakeholder collaboration, and data-driven problem-solving across cross-functional teams.